

SELECTED CCX RESULT

CCX_Results003

Fem::FemResultObjectPython

1. MODEL OVERVIEW

Project	FEMExample
Document	C:/Users/User2/Desktop/FEMExample.FCStd
Analysis	Analysis
CCX Result	CCX_Results003
PostPipeline	Pipeline_CCX_Results

2. MATERIAL PROPERTIES

Material	CalculiX-Steel
Young's modulus E	210000.0 MPa
Poisson's ratio ν	0.300
Density ρ	7900 kg/m ³
Yield strength Re	not available

3. FEM MESH PARAMETERS

Number of nodes (FEM / result)	569 / 569
Number of elements	242
Element type	C3D10
Description	tetra10 – quadratic tetrahedron
Element distribution	C3D10 × 242

4. BOUNDARY CONDITIONS

Type	Name	Reference
Fixed support	ConstraintFixed	Cube:('Face1,')
Force	ConstraintForce	Cube:('Face2,')

5. MAIN RESULTS OF SELECTED CCX RESULT

MAX. VON MISES STRESS

423.60 MPa

MAX. DEFORMATION / DISPLACEMENT

87.94 mm

MAX. PRINCIPAL STRESS

485.52 MPa

von Mises – minimum / maximum	3.492 MPa / 423.599 MPa
Displacement magnitude – minimum / maximum	0.00000 mm / 87.93699 mm
Maximum principal stress – minimum / maximum	-180.076 MPa / 485.515 MPa
Minimum principal stress – minimum / maximum	-485.698 MPa / 181.924 MPa
Safety factor $n = Re / \sigma_{VM,max}$	cannot be determined

6. CONCLUSION AND TECHNICAL ASSESSMENT

CONCLUSION

Stress: maximum von Mises stress = 423.60 MPa; maximum principal stress = 485.52 MPa.
Yield strength Re is not available; automatic safety-factor calculation was not performed.
Deformation: maximum displacement Umax = 87.937 mm.
Allowable displacement Uallow is not defined in the report configuration.
Mesh: 569 FEM/result nodes; 242 elements; C3D10.
Model validity: local maxima at supports, sharp edges or point load applications may be singularities and require separate engineering verification.

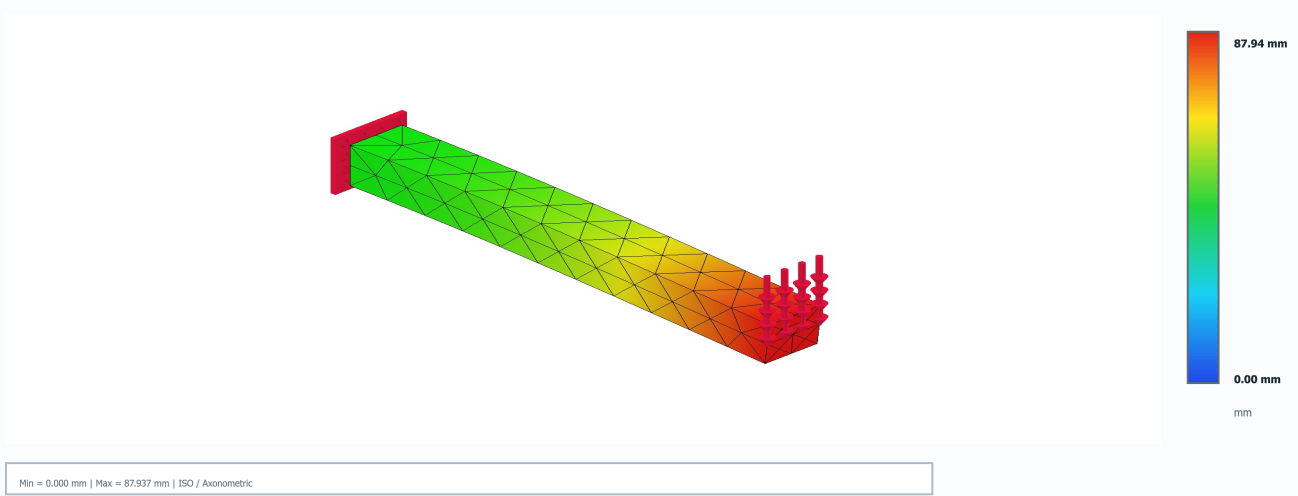
7. IDENTIFICATION AND TRACEABILITY

CCX Result object	CCX_Results003
Internal Name	CCX_Results
Result Mesh	CCX_Results_Mesh003
Object type	Fem::FemResultObjectPython
Report generation date	02.10.2026 06:25

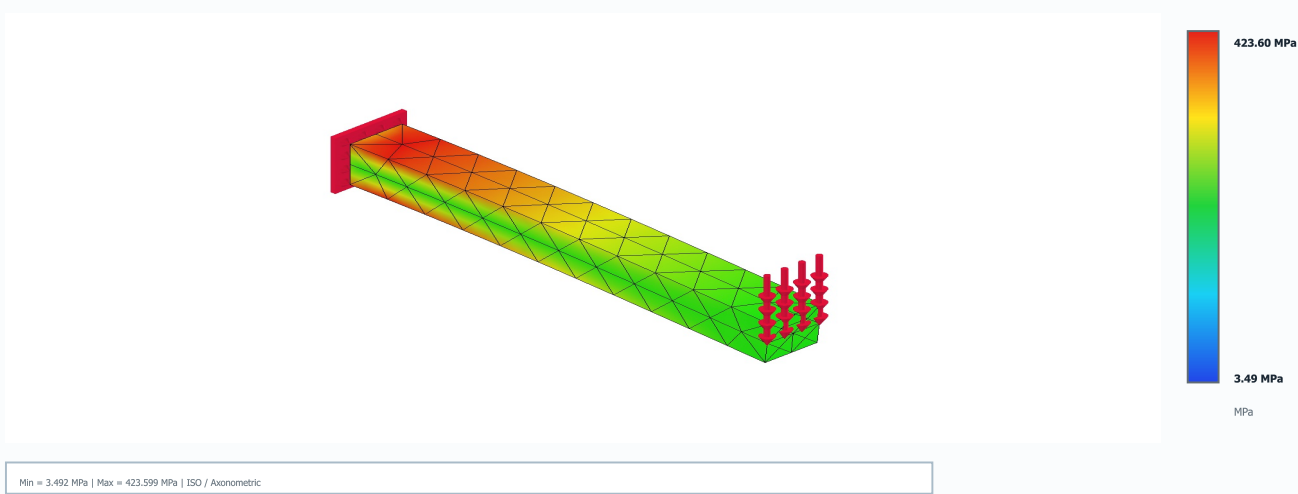
GRAPHICAL RESULTS – CCX RESULT

Result: CCX_Results003 | Project: FEMExample

CCX Result - Displacement magnitude (ISO view)



CCX Result - von Mises Stress (ISO view)



CCX Result - Maximum principal stress (ISO view)

